Kerian Thuillier

Chargé de Recherche CNRS
PhD in Computer Science

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Research Profile

Keywords: optimization modulo theory (OMT) – machine learning – dynamic systems – formal methods – bioinformatics

My research lies at the intersection of **formal methods, hybrid optimization, and machine learning**. Specifically, I focus on solving **inverse problems in complex dynamic systems** inspired by biology, using SMT-based and machine learning frameworks. My work addresses the core challenges of **scalability, robustness, and explainability** within these domains through the development of **hybrid approaches** that integrate formal guarantees with the predictive power of machine learning.
*I am committed to **open science** and to ensuring the **reproducibility** of my results.*

Current Position

***Chargé de Recherche* at CNRS** Nov. 2026 – *today*
Université Paris-Saclay, CNRS, ENS Paris-Saclay, LMF (UMR 9021), Gif-sur-Yvette, France

Background & Education

Postdoctoral Researcher Dec. 2024 – Nov. 2026
Université Paris-Saclay – PEPR IA SAIF / LMF (UMR 9021)

`Advisors:` Benedikt Bollig, Matthias Függer, Thomas Nowak

`Research topic:` Scalable and explainable machine-learning methods for parameter estimation in chemical reaction networks (CRN) modeling bacterial growth in bioreactors.

PhD in Computer Science Oct. 2021 – Sep. 2024
Université de Rennes – CNRS / IRISA (UMR 6074)

`Thesis defended:` 27 September 2024

`Title:` *Hybrid Satisfiability Methods for the Inference of Boolean Regulatory Rules Controlling Metabolic Networks*

`Supervisors:` Anne Siegel (CNRS / IRISA - Rennes), Loïc Paulevé (CNRS / LaBRI - Bordeaux)

`Jury:` François Fages (Centre INRIA Saclay), Simon de Givry (INRAE Toulouse), Emmanuelle Becker (Univ. Rennes / IRISA - Rennes), Misbah Razzaq (INRAE Tours), Laurent Tournier (INRAE Jouy-en-Josas)

Research-Oriented Master's Degree in Computer Science 2018 – 2021
École Normale Supérieure de Rennes (ENS Rennes)

Master's Degree in Computer Science 2019 – 2021
Université de Rennes 1

`Dissertation:` *Inferring Boolean regulatory rules controlling hybrid biological models*

Bachelor's Degree in Computer Science 2015 – 2019
Université de Rennes 1 (third year at ENS Rennes)

Qualifications

Qualification for *Maître de Conférences*, CNU Section 27 — Valid 2025–2029

Publications

All scripts and datasets required to reproduce results are publicly available on GitHub.

Peer-Reviewed Journal Articles

- **M. Függer, T. Nowak, and K. Thuillier** (2025). *Distributed computing inspired by biology.* **Seminars in Cell & Developmental Biology**, 175.

- **K. Thuillier, C. Baroukh, A. Bockmayr, L. Cottret, L. Paulevé, and A. Siegel** (2022). *MERRIN: Metabolic Regulation Rule Inference from time series data.* **Oxford Bioinformatics**.
  Presented at **European Conference on Systems Biology** (ECCB)

Peer-Reviewed International Conferences

- **K. Thuillier, A. Siegel, and L. Paulevé** (2024). *CEGAR-Based Approach for Solving Combinatorial Optimization Modulo Quantified Linear Arithmetics Problems.* **AAAI Conference on Artificial Intelligence**.

- **K. Thuillier, C. Baroukh, A. Bockmayr, L. Cottret, L. Paulevé, and A. Siegel** (2021). *Learning Boolean Controls in Regulated Metabolic Networks: A Case-Study.* **Computational Methods in Systems Biology (CMSB)**.

Invited Talks & Oral Presentations

Invited Talks

- *PEPR IA - SAIF Annual Meeting*, CEA List, Gif-sur-Yvette — May. 2026
  20min, `Topic:` Taylor-based ODE Solving for Parametrization of Chemical Reaction Network

- *Agglometabiont Days*, IRISA (UMR 6074), Rennes — Dec. 2025
  20min, `Topic:` Regulatory controls of bacteria metabolism

- Laboratoire Méthodes Formelles (LMF, UMR 9021), Gif-sur-Yvette — Dec. 2024
  1h, `Topic:` talking about my PhD results

- Laboratoire d'Informatique et des Systèmes (LIS, UMR 7020), Marseille — Dec. 2024
  1h, `Topic:` talking about my PhD results

- *Knowledge Representation and Reasoning* team, University of Potsdam, Germany — 2023
  1h, *Weekly seminar*, `Topic:` Solving quantified linear constraints in ASP

- *ASP days*, LaBRI (UMR 5800), Bordeaux — Nov. 2022
  20min, `Topic:` Solving quantifier-free linear constraints in ASP

Oral Presentations on Extended Abstracts

- Workshop *Modélisation du Métabolisme*, Toulouse — 2024
  Inférence de règles de régulation métabolique à partir de séries temporelles

- Session *Operational Research in Bioinformatics*, ROADEF, Rennes — 2023
  Satisfaction of Hybrid Optimization Problems Applied to Bioinformatics

- *GT-Bioss* national days, Nantes 2022
  MERRIN: MEtabolic Regulation Rule INference from time series data
- Session *Optimisation Combinatoire pour la Bioinformatique*, ROADEF, Montpellier 2020
  A MILP Approach for Genome Haplotyping
- Session *Optimisation Combinatoire pour la Bioinformatique*, ROADEF, Montpellier 2020
  A MILP Approach for the Metabolic Network Completion Problem

Softwares and Research Artifacts

All softwares and datasets are open-sources, maintained and available on GitHub.

- `maktorch [Soon publicly available]` Software
  PyTorch module; Differentiable ODE solver for mass-action kinetics based on local Taylor-series expansions
- `merrinasp` Software
  ASP-based solver for combinatorial optimization modulo quantified linear arithmetic (`OMT+qLRA`)
- `merrin` Software
  Tool for the inference of regulatory rules controlling the metabolism
- **Benchmarks of `OMT+qLRA` problems** `[Zenodo]` Benchmark
  First benchmark for this class of problems, inspired from Systems Biology
- **Regulated metabolic models of *E.coli*** Biological Models
  Models reconstructed from litteratures, no previous computational models available

Preliminary Patent Application

- **B. Bollig, M. Függer, T. Nowak, and K. Thuillier** (2025). `Under review`
  IPC: `G06N 20/00` and `C12Q 1/00`
  `Title:` *Parametrization of biochemical digital twins on timed data with deep neural networks*
  `Content:` Computational method to decrease memory usage and computation times for training physics-informed neural networks (PINNs) in the context of parameter estimation of powerlaw ODE systems modeling bacterial growth dynamics. The core of the method relies on Taylor series approximations of ODEs to simulate ODE dynamics.

Outreach

Participant and award recipient of *Sciences en Cour[t]s* 2022
`5-minute short film:` Patatogène
`Awards`: *Audience Award, Best Screenplay Award, Science Outreach Award (tie)*

Supervision

Supervision of Master 1's internship May – Jul. 2023
Master 1 student in Bioinformatics, Université de Rennes
`Topic:` Inference of metabolic regulatory rules using Answer Set Programming